

TRSI - Homework 2

Papers

1. Zendel, Oliver, Markus Murschitz, Marcel Zeilinger, Daniel Steininger, Sara Abbasi and Csaba Beleznai. "RailSem19: A Dataset for Semantic Rail Scene Understanding." 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW) (2019): 1221-1229.

Main contribution: A robust dataset for semantic rail scenes understanding (specific images taken by trams / trains and segmented to try to achieve autonomous driving for trains / trams) with 8500 images.

Results:

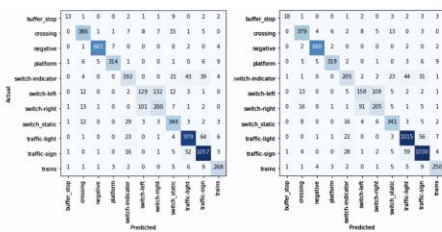


Figure 5. Confusion matrices for the all-classes classifier experiment on RailSem19 data; left: validation dataset, right: test dataset.

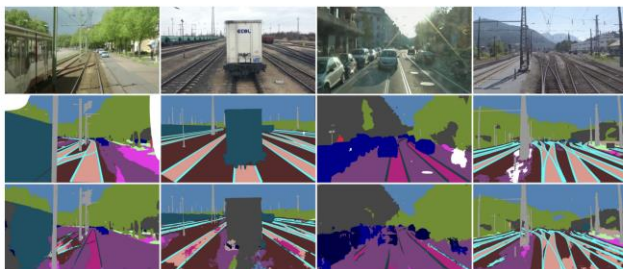


Figure 6. Example results from FRRNB experiment 6.2 for frames taken from RailSem19's validation subset (input images, dense ground truth, and prediction from FRRNB model). For the color legend, see Table 2.

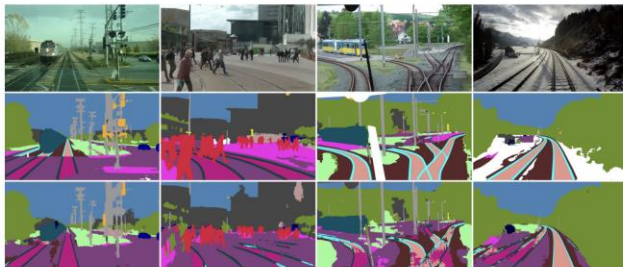


Figure 7. Further example results from FRRNB experiment 6.2 for frames taken from RailSem19's test subset.

Approach limits: Even though the numbers show that the models did great on this dataset, the authors show their limitations, the fact that the trained algorithm doesn't make great predictions on the validation dataset, mainly because of the lack of data.

2. Rombach, Robin, A. Blattmann, Dominik Lorenz, Patrick Esser and Björn Ommer. “High-Resolution Image Synthesis with Latent Diffusion Models.” 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2021): 10674-10685.

Main contribution: The diffusion models, a new approach at the time, meant to generate realistic images, the data generated by their models was by far one of the best at that time, to this date, there is no approach that made better results.

Results:

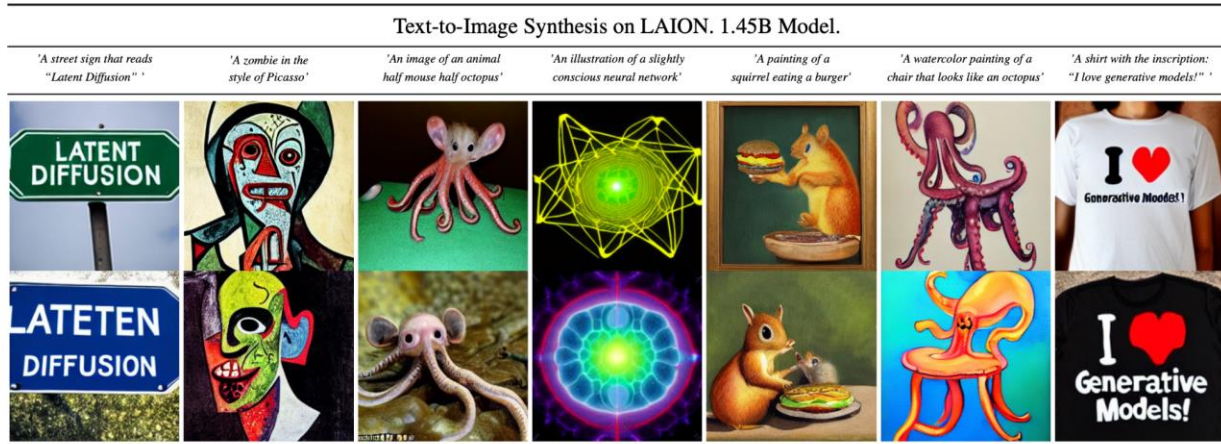


Figure 5. Samples for user-defined text prompts from our model for text-to-image synthesis, *LDM-8 (KL)*, which was trained on the LAION [78] database. Samples generated with 200 DDIM steps and $\eta = 1.0$. We use unconditional guidance [32] with $s = 10.0$.

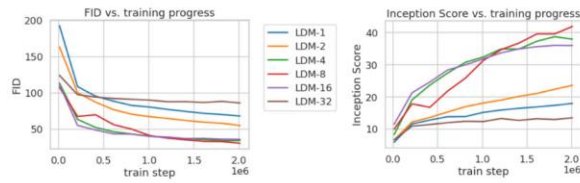


Figure 6. Analyzing the training of class-conditional *LDMs* with different downsampling factors f over 2M train steps on the ImageNet dataset. Pixel-based *LDM-1* requires substantially larger train times compared to models with larger downsampling factors

CelebA-HQ 256 × 256				FFHQ 256 × 256			
Method	FID ↓	Prec. ↑	Recall ↑	Method	FID ↓	Prec. ↑	Recall ↑
DC-VAE [63]	15.8	-	-	ImageBART [21]	9.57	-	-
VQGAN+T [23] (k=400)	10.2	-	-	U-Net GAN (+aug) [77]	10.9 (7.6)	-	-
PGGAN [39]	8.0	-	-	UDM [43]	5.54	-	-
LSGM [59]	7.22	-	-	StyleGAN [41]	4.16	0.71	0.46
UDM [43]	7.16	-	-	ProjectedGAN [76]	3.08	0.65	0.46
<i>LDM-4</i> (ours, 500-s)	5.11	0.72	0.49	<i>LDM-4</i> (ours, 200-s)	4.98	0.73	0.50
LSUN-Churches 256 × 256				LSUN-Bedrooms 256 × 256			
Method	FID ↓	Prec. ↑	Recall ↑	Method	FID ↓	Prec. ↑	Recall ↑
DDPM [30]	7.89	-	-	ImageBART [21]	5.51	-	-
ImageBART [21]	7.32	-	-	DDPM [30]	4.9	-	-
PGGAN [39]	6.42	-	-	UDM [43]	4.57	-	-
StyleGAN [41]	4.21	-	-	StyleGAN [41]	2.35	0.59	0.48
StyleGAN2 [42]	3.86	-	-	ADM [15]	1.90	0.66	0.51
ProjectedGAN [76]	1.59	0.61	0.44	ProjectedGAN [76]	1.52	0.61	0.34
<i>LDM-8*</i> (ours, 200-s)	4.02	0.64	0.52	<i>LDM-4</i> (ours, 200-s)	2.95	0.66	0.48

Approach limits: Even though the results are great, we cannot have that great of a control over what this model generates, the one control that this model takes is a text prompt.

3. Zhang, Lvmin, Anyi Rao and Maneesh Agrawala. “Adding Conditional Control to Text-to-Image Diffusion Models.” 2023 IEEE/CVF International Conference on Computer Vision (ICCV) (2023): 3813-3824.

Main contribution: The ControlNet, a model that is an extension of the diffusion model, meant to help the original model with extra information about the output (canny edges, segmentation masks, etc.), similar to the Teacher-Student approach published years ago (where a smaller network - student, tries to replicate what a bigger network - teacher predicts, to make the model as small as possible).

Results:

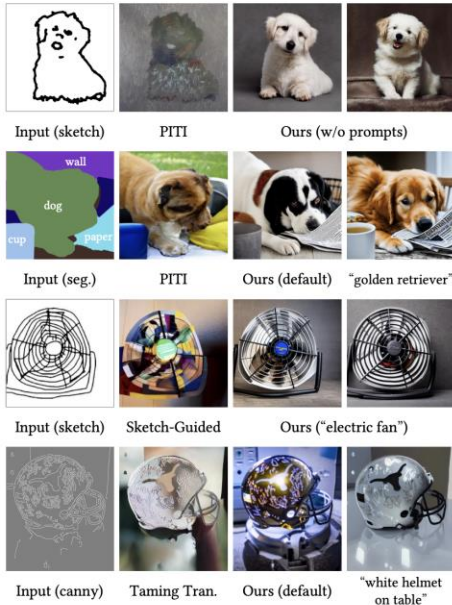


Figure 9: Comparison to previous methods. We present the qualitative comparisons to PITI [89], Sketch-Guided Diffusion [88], and Taming Transformers [19].

Method	Result Quality $\uparrow$	Condition Fidelity $\uparrow$
PITI [89](sketch)	1.10 ± 0.05	1.02 ± 0.01
Sketch-Guided [88] ($\beta = 1.6$)	3.21 ± 0.62	2.31 ± 0.57
Sketch-Guided [88] ($\beta = 3.2$)	2.52 ± 0.44	3.28 ± 0.72
ControlNet-lite	3.93 ± 0.59	4.09 ± 0.46
ControlNet	4.22 ± 0.43	4.28 ± 0.45

Table 1: Average User Ranking (AUR) of result quality and condition fidelity. We report the user preference ranking (1 to 5 indicates worst to best) of different methods.

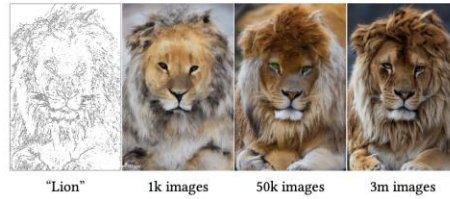


Figure 10: The influence of different training dataset sizes. See also the supplementary material for extended examples.



Figure 11: Interpreting contents. If the input is ambiguous and the user does not mention object contents in prompts, the results look like the model tries to interpret input shapes.



Figure 12: Transfer pretrained ControlNets to community models [16, 61] without training the neural networks again.

Method	FID $\downarrow$	CLIP-score $\uparrow$	CLIP-aes. $\uparrow$
Stable Diffusion	6.09	0.26	6.32
VQGAN [19](seg.)*	26.28	0.17	5.14
LDM [72](seg.)*	25.35	0.18	5.15
PITI [89](seg.)	19.74	0.20	5.77
ControlNet-lite	17.92	0.26	6.30
ControlNet	15.27	0.26	6.31

Table 3: Evaluation for image generation conditioned by semantic segmentation. We report FID, CLIP text-image score, and CLIP aesthetic scores for our method and other baselines. We also report the performance of Stable Diffusion without segmentation conditions. Methods marked with “*” are trained from scratch.

Approach limits: For the proposed task (augmenting the RailSem19 dataset with generated images), using only this model, we cannot generate some great images, because this model was trained on little to none images in the context of trains / trams.

4. Sushko, Vadim, Dan Zhang, Juergen Gall and Anna Khoreva. “One-Shot Synthesis of Images and Segmentation Masks.” 2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (2022): 6274-6283.

Main contributions: The OSMIS model, a GAN-based model that generates a segmentation mask and a realistic image at the same time, based on a context, which could be perfect for our task.

Results:

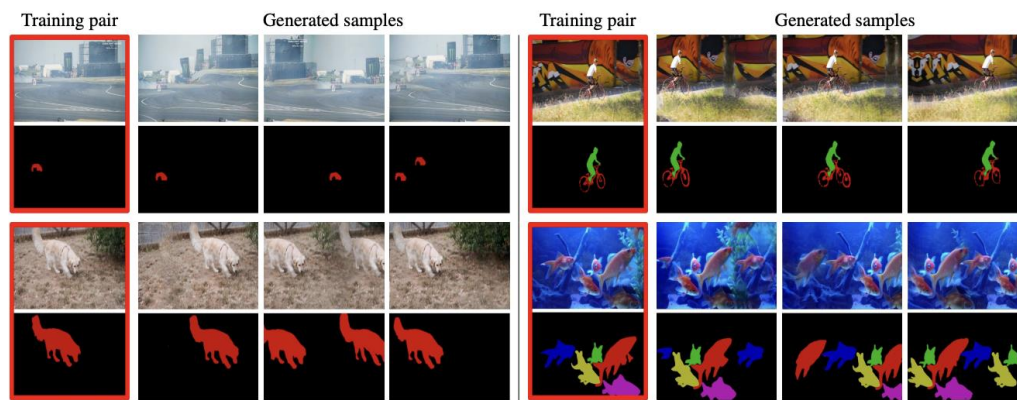


Figure 4. Qualitative results of OSMIS on DAVIS [26]. Given a single image-mask pair for training, our model achieves high-fidelity image synthesis with a high structural diversity, changing the positions of objects or editing the layout of backgrounds. For each synthesized image, it produces segmentation masks that accurately annotate the generated content. Training pairs are shown in red frames.

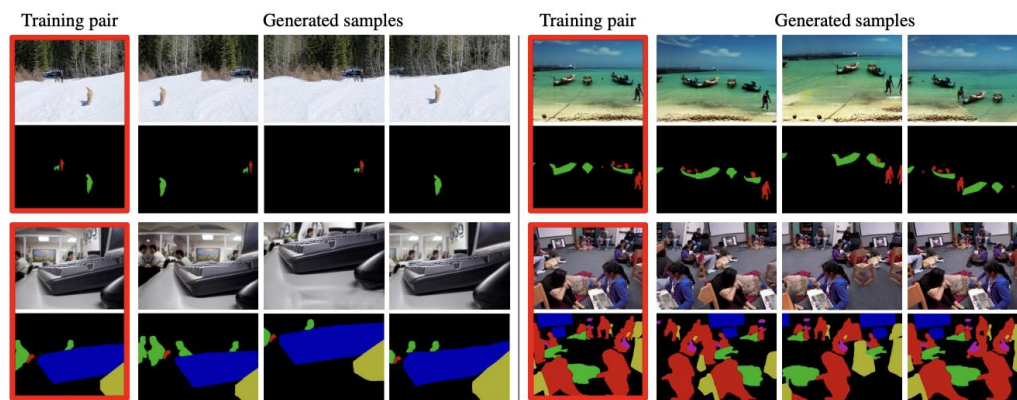


Figure 5. Qualitative results of OSMIS on COCO [20]. OSMIS successfully deals with different scene types and annotation styles. For example, it achieves high quality and diversity for both indoor and outdoor scenes, or sparse and dense annotations of foreground objects.

Method	SIFID↓	LPIPS↑
SinGAN [30]	0.131	0.267
ConSinGAN [12]	0.103	0.296
One-Shot GAN [32]	0.091	0.347
OSMIS (ours)	0.073	0.387

Table 1. Comparison of image quality and diversity to single-image GANs on DAVIS-17. Bold denotes the best performance.

Method	SIFID↓	LPIPS↑	mIoU
DatasetGAN [38]	0.118	0.007	91.1*
SemanticGAN [18]	0.211	0.012	65.8
OSMIS (ours)	0.073	0.387	86.6

Table 2. Comparison to prior image-mask GANs on DAVIS-17. Bold denotes the best performance. Red indicates mode collapse. * Indicates manual annotation of masks for DatasetGAN training.

Mask supervision	SIFID↓	LPIPS↑	mIoU
None	0.071	0.368	-
Projection [22]	0.071	0.362	72.1
Input concat.	0.079	0.328	82.4
SemanticGAN D_m [18]	0.074	0.351	83.3
MCA (ours)	0.073	0.387	86.6

Table 3. Comparison of MCA to other mask synthesis supervision mechanisms on DAVIS-17. Red indicates decreased diversity compared to the baseline. Bold denotes the best performance.

OSMIS successfully deals with very different scene types (e.g., both indoor and outdoor scenes), supports masks with both sparse and dense object annotations (e.g., foreground objects occupying small or large image areas), and can handle masks with many objects or even separate instances of the same semantic class (e.g., fish in 4th example in Fig. 4).

Approach limits: We don't have that great of a control over the GAN, it generates the images only from the latent space, even tho the approach is the most similar to the proposed task, the authors found it good to only generate some small differences from the original images, which would not help that great to generate some images in various contexts.

Proposed solution

The proposed solution is to fine-tune the Stable Diffusion model to generate segmentation masks controlled by some prompts (structural prompts, so they would control how the image is structured), and take them as a control for the ControlNet model in addition to another prompt (environmental prompt, so some things about the environment the synthetic photo was taken, like there are snow on the tracks, there is a mountain on the background, etc.). In that way, we could have full control about the outputs of those images, so we can train a segmentation model to predict those photos in different scenarios, different environments, so a very balanced dataset.

Because we have only pairs of (segmentation mask, image), we can use a pretrained image-to-description network (BLIP2 is the SoTA at the moment) to generate some predictions so we could further train the models with this additional control, so we can give other controls at the inference step.

Motivation: The lack of data in the biggest dataset at the moment in the train / trams autonomous driving scenario (RailSem19), we need a strong model to generate images in various scenarios to be able to train segmentation models. Stable Diffusion is the state of the art model at the moment, which is trained on the LAION-5B (2.3 billion images), so it surely saw various scenarios (and it was kinda proven by the images it generates), and the solution is proposed with the hope that, by fine tuning this model, it will be able to generate various train / tram scenarios and we could later train a robust segmentation model in the proposed scenario.

Work plan

Dataset: RailSem19 (<https://www.wilddash.cc/railsem19>), obtained by emailing the owners of the dataset.

Programming language: Python

Frameworks: Pytorch, Diffusers (from huggingface, where ControlNet and Stable Diffusion are defined), OpenCV (image processing), Jupyter (notebooks), numpy (arrays manipulation to compute array operations faster than in Python), matplotlib (to visualize the data), transformers (for the BLIP2 model), sklearn (mainly used for dimensions reduction, for understanding what the models does at different steps) and pandas (to manipulate data from json files easily).

Metrics: IS (Inception Score), FID (Frechet Inception Distance) for the realism of the predicted image, and to train a segmentation model on the real data only / hybrid data (real + synthetic) to see if the performance increases.